

Nader Khalil

Software Engineer

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Professional Summary

Software engineer with practical experience in full-stack web platforms, REST APIs, and foundational LLM integrations. Proven track record implementing tool calling, structured data extraction, and role-based web applications. Combines systems fundamentals in C/C++ memory management with modern application development in Python, React, and C#.

Technical Skills

AI/Agentic

LLM API Integration, Tool Calling, Structured JSON, Prompt Engineering, MCPs, Agentic Workflows

Languages

Python, Next.js, PHP, C, C++, JavaScript, TypeScript, SQL

Backend/APIs

REST APIs, C# MinimalAPI, JSON Processing

Frontend

Next.js, React, JavaScript, TypeScript, HTML5, CSS3

Databases

PostgreSQL, MySQL, EF Core

Tools/Core

Git, Linux, OOP, Memory Management, Algorithms, Docker

Professional Experience

AI Trainer & Mentor, Department of Education and Knowledge (ADEK)

07/2026 – 08/2026

Abu Dhabi, UAE

- Delivered technical instruction on agentic workflows, prompt structuring, and autonomous tool calling for the AgenticVerse summer program.
- Mentored cohort teams building functional AI prototypes tailored to industry problem statements from ADNOC, FANR, and Khalifa Fund.
- Guided students through code debugging, API schema validation, and structured output parsing.

Software Engineering Intern, Timex IT Consulting

01/2024 – 04/2024

Abu Dhabi, UAE

- Participated in software release cycles by isolating application defects and documenting functional bugs in proprietary accounting software.
- Coordinated with developers to verify technical fixes based on defect reports before client deployment.
- Built C# MinimalAPI endpoints supporting sorting, pagination, and file handling.
- Implemented request validation and standardized HTTP response handling across CRUD services.

QA & Software Tester, 42 Abu Dhabi

01/2026

Abu Dhabi, UAE

- Tested internal student platform features for functional and UI regressions prior to deployment.
- Wrote detailed defect reproduction steps to assist core developers with debugging.

IT Support & Technical Staff, Ticketmaster

12/2025
Abu Dhabi, UAE

- Configured and deployed networked ticket-scanning hardware during high-attendance operations.
- Performed on-site hardware and connectivity troubleshooting to ensure scanning uptime.

Technical Projects

AI Appointment Scheduler

09/2026 – 10/2026

- Built and deployed a full-stack appointment scheduler with role-based access for students, faculty, and admins, using Next.js 16, TypeScript, Prisma 7, PostgreSQL + pgvector, and Auth.js.
- Implemented a RAG assistant with grounded citations and agentic tool calling: the LLM prepares bookings/cancellations, but mutations execute only through permission-checked REST APIs after explicit user confirmation, with client-supplied tool output discarded and re-executed server-side.
- Hardened and verified the system with Zod validation, per-user rate limiting, prompt-injection defenses, and 239 unit / 171 integration / 49 Playwright tests in CI, plus a Docker deployment and operations runbook.

Student-Faculty Services v2, Al Ain University, Capstone Project

- Rebuilt my 2024 senior capstone as a React 19, TypeScript, and PHP 8.4 app on MariaDB, designed for free hosting with no cron, SSH, or Node and deployed by GitHub Actions once every test passes. Includes reminders, ratings, and a 10-second undo for cancellations.
- Prevented double bookings with ordered database row locks, proven by a race test where 8 students book one slot over real HTTP, and tested with PHPUnit, Vitest, Playwright, and axe accessibility checks (WCAG 2.1 AA).

Minishell (Unix Shell Implementation), 42 Abu Dhabi

- Developed a functional command interpreter in C supporting pipelines, environment variable expansion, and stream redirection.
- Managed process lifecycles via fork(), execve(), and signal handling routines.
- Prevented memory leaks across dynamic allocations using strict Valgrind analysis.

Education & Certifications

B.Sc. in Software Engineering, Al Ain University

09/2019 – 05/2024

Computer Science Curriculum, 42 Abu Dhabi

12/2024 – Present

MOEI × 42 Abu Dhabi AgentEra AI Hackathon Certificate

AI on Watch with Datamind – RecognAIze Hackathon Certificate

Languages

Arabic

Native

English

Professional Proficiency